

ArithmeticSite

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Chapter 1

The multiplicative monoid $\mathbb{N}^\times$

The Arithmetic Site, introduced by Connes and Consani, is a geometric object — a topos equipped with a structure sheaf — whose points in the topos-theoretic sense are in natural bijection with the adèle class space of $\mathbb{Q}$. Its construction begins with a single, elementary monoid: the positive integers under multiplication. Everything else — the tropical semiring, the presheaf topos, the structure sheaf, and ultimately the connection to the Riemann zeta function — is built on top of this one piece of data.

Definition 1.1. Let $\mathbb{N}^\times$ denote the set $\{1, 2, 3, \dots\}$ of strictly positive integers equipped with ordinary multiplication. This is a commutative monoid with unit 1. In Lean we represent this as `PNat` with its multiplicative structure, via the abbreviation `NPos`.

The unique factorisation theorem says that $\mathbb{N}^\times$ is the free commutative monoid on the set of primes: every element factors uniquely as a finite product $\prod_p p^{a_p}$. This arithmetic richness — the interplay of primes, divisibility, and factorisation — is what makes $\mathbb{N}^\times$ the right monoid for the Arithmetic Site. As we will see, it is ultimately responsible for the fact that the points of $\widehat{\mathbb{N}^\times}$ are parametrised by the adèle class space of $\mathbb{Q}$, and that the Hasse–Weil zeta function of the site recovers the Riemann zeta function.

Definition 1.2. The *classifying category* $B\mathbb{N}^\times$ of $\mathbb{N}^\times$ is the one-object category whose single object is $*$ and whose morphisms $*$ $\rightarrow$ $*$ are the elements of $\mathbb{N}^\times$, with composition given by multiplication.

Remark 1.3. The passage from a monoid M to the one-object category BM is the standard *delooping* construction. A presheaf on BM — that is, a functor $BM^{\text{op}} \rightarrow \mathbf{Set}$ — is exactly a set X equipped with a left action of M : each element $n \in M$ acts as a function $X \rightarrow X$, and the action is compatible with composition. (Left and right actions coincide here since $\mathbb{N}^\times$ is commutative.) The presheaf topos $\widehat{B\mathbb{N}^\times}$ introduced in Chapter 3 is precisely the category of such sets-with- $\mathbb{N}^\times$ -action, and it is in this topos that the Arithmetic Site lives.

Chapter 2

The tropical semiring $\bar{\mathbb{N}}$

The tropical semiring $\bar{\mathbb{N}}$ is the second fundamental ingredient of the Arithmetic Site. It plays the role of the *structure sheaf*: as Connes and Consani put it, $\bar{\mathbb{N}}$ is the semiring on which $\mathbb{N}^\times$ acts by multiplication, and it is this action that makes $\bar{\mathbb{N}}$ into a sheaf of semirings over the topos $\widehat{B\mathbb{N}^\times}$. The arithmetic of $\bar{\mathbb{N}}$ is characteristic-1 arithmetic — addition is idempotent — which is the tropical, or min-plus, shadow of ordinary arithmetic. Connes and Consani define $\bar{\mathbb{N}}$ explicitly as $(\mathbb{N} \cup \{+\infty\}, \inf, +)$, confirming the min-plus convention: addition is min, with $+\infty$ as the additive identity. When elements of $\bar{\mathbb{N}}$ are written as exponents q^x for $q \in (0, 1)$, the addition becomes max on values (since smaller exponent means larger value), which is why the stalks in the paper appear in max-plus form as $[0, 1]_{\max}$ and $\mathbb{R}_{\max}$.

Definition 2.1. The *tropical semiring* $\bar{\mathbb{N}}$ is the set $\mathbb{N} \cup \{+\infty\}$ equipped with:

- *Tropical addition*: $a \oplus b := \min(a, b)$, with identity element $+\infty$.
- *Tropical multiplication*: $a \odot b := a + b$ (ordinary addition), with identity element 0.

In Lean, $\bar{\mathbb{N}}$ is represented as `Tropical (WithTop ℕ)`, via the abbreviation `NBar`. The zero element is `trop ⊤` (i.e. $+\infty$) and the multiplicative identity is `trop 0`.

Lemma 2.2. $\bar{\mathbb{N}}$ is a commutative semiring.

Proof. This is the Mathlib instance `CommSemiring (Tropical (WithTop ℕ))`, found by `inferInstance`. □

Lemma 2.3. Tropical addition on $\bar{\mathbb{N}}$ is idempotent: $a \oplus a = a$ for all $a \in \bar{\mathbb{N}}$.

Proof. $a \oplus a = \min(a, a) = a$. In Lean this follows from `Tropical.add_eq_left`. □

Remark 2.4. Idempotency of addition — equivalently, the identity $1 \oplus 1 = 1$ — is the defining feature of *characteristic 1* arithmetic. It is what places $\bar{\mathbb{N}}$ in the same conceptual family as finite fields $\mathbb{F}_p$ (characteristic p), with min playing the role that $+\text{ mod } p$ plays there. The scaling endomorphisms defined below are the characteristic-1 analogue of the Frobenius endomorphism $x \mapsto x^p$ of $\mathbb{F}_p$.

Definition 2.5. For each $n \in \mathbb{N}^\times$, the *scaling endomorphism* $\varphi_n : \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$ is defined by

$$\varphi_n(x) := nx \quad (x \in \mathbb{N}), \quad \varphi_n(+\infty) := +\infty,$$

where nx denotes ordinary multiplication in $\mathbb{N}$. Equivalently, $\varphi_n(x) = x^{\odot n}$, the n -fold tropical product of x with itself. In Lean, φ_n is defined as a semiring endomorphism $\mathbb{N}\text{Bar} \rightarrow^{+*} \mathbb{N}\text{Bar}$ via `scalingEndomorphism`.

Lemma 2.6. *For each $n \in \mathbb{N}^\times$, the map φ_n is a semiring endomorphism of $\bar{\mathbb{N}}$: it preserves both operations and both identities.*

Proof. Preservation of $+\infty$ and of tropical multiplication are immediate. Preservation of tropical addition uses $n \cdot \min(a, b) = \min(na, nb)$ for $n > 0$, which holds because scaling by n is order-preserving on $\mathbb{N} \cup \{+\infty\}$. In Lean this is the `map_add'` field of `scalingEndomorphism`, proved via monotonicity of `nsmul` and a case split on `le_total`. $\square$

Lemma 2.7. *The scaling endomorphisms compose multiplicatively:*

$$\varphi_m \circ \varphi_n = \varphi_{mn}$$

for all $m, n \in \mathbb{N}^\times$. In particular, the map $n \mapsto \varphi_n$ is a monoid homomorphism from $(\mathbb{N}^\times, \times)$ into the semiring endomorphism monoid $\text{End}_{\text{Semiring}}(\bar{\mathbb{N}})$.

Proof. $(\varphi_m \circ \varphi_n)(x) = m(nx) = (mn)x = \varphi_{mn}(x)$. In Lean this is `scalingEndomorphism_comp`, proved via `pow_mul`. $\square$

Remark 2.8. The Frobenius correspondences $\Psi(\lambda)$, parametrised by $\lambda \in \mathbb{Q}_+^\times$, are a more refined object: they live in the square $\widehat{B\mathbb{N}^{\times 2}}$ of the Arithmetic Site and are defined via the action of $\mathbb{N}^\times \times \mathbb{N}^\times$ on $\bar{\mathbb{N}} \otimes_{\mathbb{B}} \bar{\mathbb{N}}$. They are introduced in Chapter 7.

Chapter 3

The presheaf topos $\widehat{\mathbb{N}^\times}$

The topos underlying the Arithmetic Site is the simplest kind: a presheaf topos. Its objects are sets equipped with a left $\mathbb{N}^\times$ -action by arbitrary functions — not necessarily bijections, since $\mathbb{N}^\times$ is a monoid rather than a group — and its morphisms are $\mathbb{N}^\times$ -equivariant maps between them. The multiplicative structure of $\mathbb{N}^\times$ is entirely encoded in the composition of these actions: if n and m act on a set X , then nm acts as the composite $x \mapsto n \cdot (m \cdot x)$.

Connes and Consani note that $\widehat{\mathbb{N}^\times} \simeq \text{Sh}(\mathbb{N}^\times, J)$, where J is the *chaotic topology* on $\mathbb{N}^\times$ — the topology in which every presheaf is already a sheaf. This means there are no genuine gluing conditions to check: the topos is simply the presheaf category, with no further data. The covering families arising from factorizations of integers, foreshadowed in Chapter 1, reflect a finer site structure that one *can* put on $\mathbb{N}^\times$; the Arithmetic Site itself uses the coarser chaotic topology, where that combinatorial structure is not needed.

Definition 3.1. The *presheaf topos* $\widehat{\mathbb{N}^\times}$ is the category of functors $B\mathbb{N}^{\times\text{op}} \rightarrow \mathbf{Set}$, i.e. the category of sets equipped with a left $\mathbb{N}^\times$ -action. In Lean this is `Action (Type*) (Multiplicative ℕ+)`.

Lemma 3.2. $\widehat{\mathbb{N}^\times}$ is a Grothendieck topos.

Proof. Any presheaf category on a small category is a Grothendieck topos. The site is $(\mathbb{N}^\times, J)$ where J is the chaotic topology. $\square$

Remark 3.3. The points of $\widehat{\mathbb{N}^\times}$ — geometric morphisms from $\mathbf{Set}$ to $\widehat{\mathbb{N}^\times}$ — are the deep link to number theory. Connes and Consani prove (Theorem 2.2 of their 2014 paper) that the category of points of $\widehat{\mathbb{N}^\times}$ is canonically equivalent to the category of totally ordered groups isomorphic to non-trivial subgroups of $(\mathbb{Q}, \mathbb{Q}_+)$. The space of isomorphism classes of such points is canonically isomorphic to the double quotient

$$\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*,$$

where $\mathbb{A}^f$ is the ring of finite adèles of $\mathbb{Q}$ and $\widehat{\mathbb{Z}}^* = \widehat{\mathbb{Z}}^\times$ is the group of units of the profinite completion of $\mathbb{Z}$. This quotient is a piece of the adèle class space of $\mathbb{Q}$, the same space that appears in Connes’ noncommutative-geometric approach to the Riemann hypothesis. The structure sheaf $\mathcal{O} = \widehat{\mathbb{N}}$ then equips this space with characteristic-1 geometry, whose Hasse–Weil zeta function recovers the Riemann zeta function.

Chapter 4

The structure sheaf

In classical algebraic geometry, a *ringed space* is a topological space equipped with a sheaf of commutative rings — the *structure sheaf* — whose stalks encode the local algebra of the space. The Arithmetic Site is a *ringed topos* in the same spirit: the role of the topological space is played by the presheaf topos $\widehat{\mathbb{N}^\times}$, and the role of the structure sheaf is played by $\bar{\mathbb{N}}$, viewed as a semiring object living inside that topos.

An object of $\widehat{\mathbb{N}^\times}$ is a set equipped with a left $\mathbb{N}^\times$ -action, so for $\bar{\mathbb{N}}$ to be a semiring *object* of $\widehat{\mathbb{N}^\times}$ the action must be by semiring endomorphisms: each scaling map φ_n must preserve both tropical operations and both identities. This is exactly the content of Lemma 2.6, which is therefore the key structural fact underlying the definition below.

Definition 4.1. The *structure sheaf* $\mathcal{O}$ of the Arithmetic Site is the object of $\widehat{\mathbb{N}^\times}$ whose underlying set is $\bar{\mathbb{N}}$ and whose $\mathbb{N}^\times$ -action is given by the scaling endomorphisms: $n \cdot x := \varphi_n(x) = nx$.

Lemma 4.2. The assignment $n \mapsto \varphi_n$ defines a monoid homomorphism $(\mathbb{N}^\times, \times) \rightarrow \text{End}_{\text{Semiring}}(\bar{\mathbb{N}})$, making $\bar{\mathbb{N}}$ into a semiring object of $\widehat{\mathbb{N}^\times}$ with $\mathbb{N}^\times$ acting by multiplication.

Chapter 5

The Arithmetic Site

Definition 5.1. The *Arithmetic Site* is the pair $(\widehat{\mathbb{N}^\times}, \mathcal{O})$ consisting of the presheaf topos $\widehat{\mathbb{N}^\times}$ together with its structure sheaf $\mathcal{O}$.

Chapter 6

Points of the topos

Definition 6.1. A *point* of a topos $\mathcal{E}$ is a geometric morphism $p : \mathbf{Set} \rightarrow \mathcal{E}$, i.e. an adjoint pair $p^* \dashv p_*$ where p^* is left exact.

Theorem 6.2. *The points of $\widehat{\mathbb{N}^\times}$ are in natural bijection with the adèle class space $\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*$.*

Proof. (sorry)

□

Chapter 7

The Frobenius correspondences

Definition 7.1. For each $\lambda \in \mathbb{Q}_+^\times$, the *Frobenius correspondence* Φ_λ is a correspondence on $\widehat{\mathbb{N}^\times} \times \widehat{\mathbb{N}^\times}$ arising from the scaling action of $\mathbb{Q}_+^\times$ on the adèle class space.

Theorem 7.2. *The Frobenius correspondences realize the scaling action of $\mathbb{Q}_+^\times$ on the adèle class space at the level of the topos $\widehat{\mathbb{N}^\times}$.*

Proof. (sorry)

□

Chapter 8

Connection to the Riemann zeta function

Theorem 8.1. *The Hasse–Weil zeta function of the Arithmetic Site recovers the Riemann zeta function $\zeta(s)$.*

Proof. (sorry)

□